Supplementary Information for "Certifying threshold management in one dimension: closed-loop robust kernels for the Northern cod limit reference point and the Edwards Aquifer index well"

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This supplement contains the locked data panels, the full kernel grids, the identification and stochastic layers, the Edwards audits, the verification summary, the two frozen protocols and a symbol table. Every table is generated by `work/make_tables.py` from `work/results.json`, the output of the re-derivation core `work/p09_core.py`; no number is typed by hand. Section, theorem, table and figure numbers of the form "Section 2" or "Theorem 2.3" refer to the main text. Main-text tables are cited as Tables 1–7.

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S1 Locked data panels and residuals

Tables S1 and S2 print the two panels exactly as read by the core, with the one-step residuals of the registered maps. The cod panel is the NCAM M-shift spawning stock biomass of the 2016 assessment (Table A2, 1983–2015) with the Schijns et al. (2021) annual landings; the fit window is 1983–2007 (24 transitions) and the audit window 2008–2015 (8 transitions). The Edwards panel is the J-17 annual mean elevation, the USGS total recharge and the Edwards Aquifer Authority's total well pumping, 1934–2023; the fit window is 1935–1990 (56 transitions) and the audit window 1991–2023 (33 transitions). The 2024 reassessment series (xteNCAM, 1954–2024) is used only for the recent-window checks of Section S6 and is not pooled with the 2016 series.

Table S1: Northern cod data (NCAM 2016 Table A2 SSB; Schijns et al. 2021 landings) and one-step residuals $e_{t+1} = S_{t+1} - [S_t + g(S_t) - C_t]$ of the registered source-year fit (residual listed under its destination year).

year	SSB (kt)	catch (kt)	residual (kt)	role
1983	841.08	232.345	–	–
1984	863.23	232.471	88.78	fit
1985	902.94	231.293	103.01	fit
1986	836.00	266.713	-10.90	fit
1987	940.75	239.924	206.55	fit
1988	886.40	268.677	4.67	fit
1989	921.66	253.990	131.20	fit
1990	861.92	219.452	16.18	fit
1991	734.51	172.012	-76.93	fit
1992	381.95	40.956	-328.97	fit
1993	101.05	11.392	-323.51	fit
1994	30.55	1.314	-82.56	fit
1995	9.68	0.413	-26.75	fit
1996	16.05	1.875	4.49	fit
1997	20.57	0.877	2.61	fit
1998	28.25	4.507	3.70	fit
1999	34.59	8.526	4.19	fit
2000	34.42	5.430	0.22	fit
2001	29.50	6.969	-7.59	fit
2002	23.87	4.249	-5.61	fit
2003	22.07	0.994	-3.18	fit
2004	20.07	0.649	-6.21	fit
2005	25.18	1.331	1.02	fit
2006	40.83	2.701	11.05	fit
2007	81.10	2.931	33.38	fit
2008	106.65	3.385	9.58	OOS
2009	104.56	3.116	-23.43	OOS
2010	96.91	2.962	-28.78	OOS
2011	90.56	3.770	-25.90	OOS
2012	112.12	3.871	4.27	OOS
2013	169.17	4.506	34.96	OOS
2014	250.12	4.870	46.74	OOS
2015	298.65	4.436	-2.88	OOS

Table S2: Edwards Aquifer annual panel 1934–2023 (J-17 mean head, ft AMSL; total recharge and well pumping, 10^3 acre-ft) and residuals of the registered affine fit (destination year).

year	H	R	P	residual (ft)	role
1934	670.41	179.6	101.9	–	–
1935	673.68	1258.2	103.7	-12.00	fit
1936	680.16	909.6	112.7	-0.80	fit
1937	678.27	400.7	120.2	2.79	fit
1938	676.86	432.7	120.1	2.15	fit
1939	670.20	399.0	118.9	-2.83	fit
1940	665.19	308.8	120.1	-1.05	fit
1941	676.37	850.7	136.8	3.61	fit
1942	676.45	557.8	144.6	1.37	fit
1943	673.14	273.1	149.1	3.78	fit
1944	672.20	560.9	147.3	-0.45	fit
1945	675.22	527.8	153.3	4.10	fit
1946	673.16	556.1	155.0	-0.73	fit
1947	672.02	422.6	167.0	2.65	fit
1948	659.35	178.3	168.7	-4.27	fit
1949	662.35	508.1	179.4	1.95	fit

(continued)

year	H	R	P	residual (ft)	role
1950	659.48	200.2	193.8	3.35	fit
1951	648.35	139.9	209.7	-3.98	fit
1952	642.63	275.5	215.4	-3.93	fit
1953	642.87	167.6	229.8	3.12	fit
1954	635.80	162.1	246.2	-3.54	fit
1955	630.84	192.0	261.0	-3.40	fit
1956	623.15	43.7	321.1	-2.74	fit
1957	637.12	1142.6	237.3	-7.21	fit
1958	666.79	1711.2	219.3	0.25	fit
1959	672.01	690.4	234.5	4.00	fit
1960	671.70	824.8	227.1	-3.07	fit
1961	674.83	717.1	228.2	2.46	fit
1962	663.47	239.4	267.9	-0.64	fit
1963	651.89	170.7	276.4	-2.13	fit
1964	648.08	413.2	260.2	-2.58	fit
1965	659.63	623.5	256.1	7.53	fit
1966	659.44	615.2	255.9	-1.12	fit
1967	644.82	466.5	341.3	-10.22	fit
1968	670.11	884.7	251.7	15.14	fit
1969	664.37	610.5	307.5	-2.45	fit
1970	665.80	661.6	329.4	2.88	fit
1971	651.61	925.3	406.8	-15.41	fit
1972	668.01	756.4	371.3	13.91	fit
1973	681.95	1486.5	310.4	-0.59	fit
1974	678.24	658.5	377.4	3.63	fit
1975	679.33	973.0	327.8	-0.17	fit
1976	678.13	894.1	349.5	-0.00	fit
1977	686.51	952.0	380.6	9.01	fit
1978	673.21	502.5	431.8	-0.17	fit
1979	682.39	1117.8	391.5	5.59	fit
1980	663.33	406.4	491.1	-3.38	fit
1981	674.41	1448.4	387.1	-1.71	fit
1982	665.47	422.4	453.1	3.32	fit
1983	657.33	420.1	418.5	0.90	fit
1984	638.80	197.9	529.8	-3.98	fit
1985	659.31	1003.3	522.5	14.18	fit
1986	666.77	1153.7	429.3	0.70	fit
1987	684.24	2003.6	364.1	-6.10	fit
1988	664.63	355.5	540.0	-1.07	fit
1989	644.14	214.4	542.4	-4.06	fit
1990	645.84	1123.2	489.4	-6.60	fit
1991	654.82	1508.4	436.0	-8.05	OOS
1992	691.96	2485.7	327.2	-0.07	OOS
1993	683.45	447.6	407.3	6.40	OOS
1994	669.72	538.1	424.6	-2.29	OOS
1995	664.47	531.3	399.6	2.13	OOS
1996	644.74	324.3	493.6	-6.90	OOS
1997	662.82	1134.6	377.1	6.51	OOS
1998	667.63	1142.3	453.5	-0.14	OOS
1999	670.40	473.5	442.7	11.99	OOS
2000	656.25	614.5	414.8	-7.81	OOS
2001	674.83	1069.4	367.7	10.97	OOS
2002	681.69	1573.7	371.3	-5.94	OOS
2003	684.38	669.0	362.1	9.32	OOS
2004	686.42	2176.1	317.4	-21.81	OOS
2005	686.13	764.0	388.5	6.40	OOS
2006	662.81	201.6	454.5	-3.68	OOS
2007	683.93	2162.3	319.9	-7.86	OOS
2008	674.19	212.9	428.6	8.38	OOS
2009	658.79	211.0	395.7	-0.64	OOS
2010	676.13	813.5	372.6	15.58	OOS

(continued)

year	H	R	P	residual (ft)	role
2011	653.71	112.0	427.7	-4.29	OOS
2012	651.56	313.5	384.7	5.07	OOS
2013	644.41	182.7	355.8	1.28	OOS
2014	636.36	107.2	332.2	-0.60	OOS
2015	652.89	1358.1	325.2	-3.08	OOS
2016	672.31	1221.1	325.3	6.74	OOS
2017	670.42	486.6	379.2	6.46	OOS
2018	664.31	1105.0	370.6	-10.75	OOS
2019	677.73	492.0	358.6	19.05	OOS
2020	664.54	184.0	362.4	2.05	OOS
2021	662.85	244.0	326.6	7.99	OOS
2022	643.45	156.0	387.2	-6.66	OOS
2023	635.68	147.0	320.5	-1.69	OOS

S2 Cod floor classes under alternative quantile conventions

The three floor classes are the minimum, the 5th and the 10th percentile of the 24 fit residuals (linear interpolation, the registered convention). Table S3 recomputes the two percentile floors under five conventions and shows the implied constants. The q05 floor is the one sensitive to the convention: with $n = 24$ the 5th percentile falls between the two collapse residuals (-329.0 and -323.5 kt) and the third-smallest (-82.6 kt), so interpolating conventions place it anywhere in that gap and the "lower" convention returns the second-smallest residual itself. The q10 floor is stable to within 26 kt under all conventions except Weibull. None of the main-text verdicts under the worst and q10 floors depends on the convention; the q05 column of Table 2 does, and is reported for that reason.

Table S3: Cod floor classes under alternative quantile conventions ($n = 24$ fit residuals) and the implied constants.

convention	e_{q05}	e_{q10}	C^* (q10)	C_{vac} (q10)	C_{vac} (q05)
linear interpolation (declared)	-287.36	-80.87	91.59	215.22	8.72
lower order statistic	-323.51	-82.56	89.90	213.53	-27.42
inverted CDF	-323.51	-82.56	89.90	213.53	-27.42
Hazen	-325.15	-106.65	65.81	189.43	-29.06
Weibull	-327.61	-203.03	-30.57	93.05	-31.52

Sorted fit residuals (kt): -329.0, -323.5, -82.6, -76.9, -26.7, -10.9, -7.6, -6.2, -5.6, -3.2, 0.2, 1.0, 2.6, 3.7, 4.2, 4.5, 4.7, 11.0, 16.2, 33.4, 88.8, 103.0, 131.2, 206.5. The two collapse residuals (destination years 1992 and 1993) are the only ones below -172.5 kt; the declared q10 floor sits between the second and third order statistics, which is why the convention matters for q10 and not for q05.

S3 Cod nominal kernels: full horizon grid and rule supplies

Table S4 gives the lower boundary of the nominal closed-loop kernel of the safe set $[\theta, 10\,000]$ kt for every rule of the registered family, every floor class and the horizons $T \in \{1, 2, 3, 5, 8, 10, 15, 20, \infty\}$, computed by the exact backward recursion of Theorem 2.3 on the worst-floor map. The Stage-I rule, the cascade and the flat 60-kt cap have identical rows under every floor (Proposition 2.6), because the kernels live in the safe set, where all three take 60 kt; the two graded rules differ from them only through their higher stages above $1.25\,\theta$.

Supply on the observed states. On the 1983–2007 states the mean catch prescribed by the rules is 5.0 kt for BAU, 10.0 kt for the Stage-I rule (cut active in 20 of 24 years, 83%), 16.3 kt for the cascade, and 60 kt for the flat 60-kt cap. The staged rules therefore fail clause (b) of the

retention rule against the flat cap they match in protection: they are strictly less productive than it on the record, not more, because the record lies below the LRP in 83% of years (20 of 24), where the Stage-I rule takes nothing and the cascade takes 30, 5 or 0 kt.

Table S4: Northern cod: nominal kernel lower boundaries b_T (kt) for every policy, floor class and horizon (exact backward recursion; $\emptyset$ = empty).

policy	class	T=1	2	3	5	8	10	15	20	∞
zero catch	worst	1021.1	1141.4	1248.5	1431.6	1650.6	1770.9	2015.3	2211.8	$\emptyset$
zero catch	q05	984.6	1072.5	1150.1	1280.8	1431.6	1510.5	1657.6	1758.0	2070.9
zero catch	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6
BAU (5 kt)	worst	1025.5	1149.7	1260.4	1450.0	1677.8	1803.7	2062.3	2275.0	$\emptyset$
BAU (5 kt)	q05	989.0	1080.7	1161.8	1298.7	1457.3	1540.7	1697.8	1806.9	2219.6
BAU (5 kt)	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6
flat 60 kt	worst	1073.8	1241.7	1392.7	1656.7	1990.5	2188.8	2657.5	3177.4	$\emptyset$
flat 60 kt	q05	1037.2	1172.0	1292.4	1499.6	1751.5	1893.2	2193.4	2456.2	$\emptyset$
flat 60 kt	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6
flat 120 kt	worst	1126.8	1343.1	1539.7	1891.7	2364.6	2671.8	3562.3	5311.8	$\emptyset$
flat 120 kt	q05	1090.0	1272.7	1437.4	1727.6	2101.2	2329.0	2898.1	3619.7	$\emptyset$
flat 120 kt	q10	909.2	930.7	949.4	979.9	1012.9	1028.6	1053.9	1067.2	1082.3
flat 180 kt	worst	1180.0	1445.6	1689.6	2137.6	2781.2	3245.5	5079.9	$\emptyset$	$\emptyset$
flat 180 kt	q05	1143.1	1374.4	1585.3	1965.9	2487.3	2836.5	3932.1	6937.9	$\emptyset$
flat 180 kt	q10	961.5	1028.8	1088.0	1186.9	1298.5	1355.2	1455.8	1518.7	1637.8
flat 240 kt	worst	1233.5	1549.2	1842.5	2395.6	3252.1	3953.2	9526.3	$\emptyset$	$\emptyset$
flat 240 kt	q05	1196.4	1477.2	1736.1	2215.3	2919.1	3445.4	5834.2	$\emptyset$	$\emptyset$
flat 240 kt	q10	1014.0	1127.9	1229.2	1401.9	1606.9	1718.5	1941.1	2114.0	$\emptyset$
S1 (staged 60 kt)	worst	1073.8	1241.7	1392.7	1656.7	1990.5	2188.8	2657.5	3177.4	$\emptyset$
S1 (staged 60 kt)	q05	1037.2	1172.0	1292.4	1499.6	1751.5	1893.2	2193.4	2456.2	$\emptyset$
S1 (staged 60 kt)	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6
cascade (60/30/5/0)	worst	1073.8	1241.7	1392.7	1656.7	1990.5	2188.8	2657.5	3177.4	$\emptyset$
cascade (60/30/5/0)	q05	1037.2	1172.0	1292.4	1499.6	1751.5	1893.2	2193.4	2456.2	$\emptyset$
cascade (60/30/5/0)	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6
graded 60/90	worst	1073.8	1268.6	1444.2	1754.3	2156.5	2404.9	3046.2	3934.1	$\emptyset$
graded 60/90	q05	1037.2	1198.7	1343.4	1594.6	1907.5	2089.8	2505.4	2930.8	$\emptyset$
graded 60/90	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6
graded 30/60/90	worst	1047.4	1218.2	1398.5	1715.1	2122.2	2371.8	3008.6	3871.4	$\emptyset$
graded 30/60/90	q05	1010.9	1148.7	1271.5	1534.6	1858.2	2044.7	2463.9	2884.4	$\emptyset$
graded 30/60/90	q10	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6	884.6

S4 Cod exact bounded-defect kernels

Table S5 gives the lower boundaries of the exact bounded-defect kernel of Theorem 2.7 with the declared defect $\varepsilon = 329.0$ kt on top of each floor class, at horizons $T = 1, \ldots, 20$, together with the last non-empty horizon and the last non-empty horizon of the Lipschitz certificate for comparison. Under the worst floor the recursion is $W_{t+1} = \{S \geq \theta : \; F(S) - |e_{\min}| - \varepsilon \; \geq \; b_t\}$ for a flat rule, so the $T = 1$ boundary solves $F(S) - 5 = \theta + 2|e_{\min}|$ for BAU; the boundaries grow because the map is expansive on the kernel and the kernel becomes empty when the boundary passes the 10 000-kt ceiling. The Lipschitz certificate (Proposition 2.8) inflates the threshold by $r_T = \varepsilon\,(F'(\theta)^T - 1)/(F'(\theta) - 1)$ instead and is empty two to twenty horizons earlier, because it pays for the worst excursion at every step rather than for the excursions the clamped state can make.

Table S5: Northern cod: exact bounded-defect kernel boundaries (kt) with $\varepsilon = 329.0$ kt on top of the floor class, and the last non-empty horizon T^{ex}.

policy	class	T^{ex}	T=1	2	3	5	8	10	12	15	20
zero catch	worst	12	1313.2	1705.0	2075.4	2803.6	4089.5	5449.6	8938.3	$\emptyset$	$\emptyset$
zero catch	q05	13	1275.8	1631.8	1965.6	2608.8	3673.2	4658.3	6414.4	$\emptyset$	$\emptyset$
zero catch	q10	28	1092.2	1276.9	1443.5	1737.3	2116.4	2348.3	2576.2	2932.6	3688.3
BAU (5 kt)	worst	12	1317.7	1713.8	2088.7	2827.6	4143.0	5560.0	9479.1	$\emptyset$	$\emptyset$

(continued)

policy	class	T^{ex}	T=1	2	3	5	8	10	12	15	20
BAU (5 kt)	q05	13	1280.3	1640.5	1978.7	2631.8	3720.7	4743.2	6623.6	∅	∅
BAU (5 kt)	q10	27	1096.6	1285.3	1455.7	1756.8	2147.2	2387.8	2626.1	3003.9	3834.9
flat 60 kt	worst	10	1367.2	1811.5	2236.7	3098.6	4792.5	7141.8	∅	∅	∅
flat 60 kt	q05	11	1329.7	1737.5	2124.4	2892.2	4290.4	5877.0	∅	∅	∅
flat 60 kt	q10	20	1145.3	1378.6	1591.5	1975.9	2504.2	2859.3	3247.5	3986.8	7275.0
flat 120 kt	worst	9	1421.5	1919.4	2401.7	3412.0	5673.9	∅	∅	∅	∅
flat 120 kt	q05	10	1383.8	1844.5	2286.9	3192.6	5039.7	7925.5	∅	∅	∅
flat 120 kt	q10	16	1198.6	1481.5	1742.4	2225.9	2938.1	3473.4	4147.8	5955.5	∅
flat 180 kt	worst	8	1476.1	2028.5	2570.6	3746.8	6861.6	∅	∅	∅	∅
flat 180 kt	q05	9	1438.2	1952.7	2453.0	3512.2	5996.2	∅	∅	∅	∅
flat 180 kt	q10	14	1252.1	1585.5	1896.5	2488.4	3431.5	4244.0	5507.3	∅	∅
flat 240 kt	worst	8	1530.9	2139.0	2743.6	4106.5	8724.4	∅	∅	∅	∅
flat 240 kt	q05	8	1492.9	2062.2	2623.2	3854.2	7328.4	∅	∅	∅	∅
flat 240 kt	q10	12	1305.9	1690.7	2053.9	2765.1	4004.5	5278.6	8240.7	∅	∅
S1 (staged 60 kt)	worst	10	1367.2	1811.5	2236.7	3098.6	4792.5	7141.8	∅	∅	∅
S1 (staged 60 kt)	q05	11	1329.7	1737.5	2124.4	2892.2	4290.4	5877.0	∅	∅	∅
S1 (staged 60 kt)	q10	20	1145.3	1378.6	1591.5	1975.9	2504.2	2859.3	3247.5	3986.8	7275.0
cascade (60/30/5/0)	worst	10	1367.2	1811.5	2236.7	3098.6	4792.5	7141.8	∅	∅	∅
cascade (60/30/5/0)	q05	11	1329.7	1737.5	2124.4	2892.2	4290.4	5877.0	∅	∅	∅
cascade (60/30/5/0)	q10	20	1145.3	1378.6	1591.5	1975.9	2504.2	2859.3	3247.5	3986.8	7275.0
graded 60/90	worst	10	1394.4	1865.3	2318.7	3252.8	5205.3	8551.4	∅	∅	∅
graded 60/90	q05	11	1356.7	1790.8	2205.2	3040.1	4645.0	6733.8	∅	∅	∅
graded 60/90	q10	18	1171.9	1429.9	1666.6	2099.4	2714.7	3150.9	3659.9	4776.5	∅
graded 30/60/90	worst	10	1394.4	1865.3	2318.7	3252.8	5205.3	8551.4	∅	∅	∅
graded 30/60/90	q05	11	1356.7	1790.8	2205.2	3040.1	4645.0	6733.8	∅	∅	∅
graded 30/60/90	q10	18	1145.3	1405.8	1644.2	2079.1	2694.3	3128.4	3632.3	4727.5	∅

S5 Cod proportional family and finite-duration floors

Table S6 evaluates proportional rules $C = \phi\, g(S)$ under the q10 floor by Corollary 2.4: the kernel is the whole safe set iff $\phi \le \phi^*_{\mathrm{safe}} = 1 - |e|/g(\theta)$, non-empty iff $\phi \le 1 - |e|/g_{\max}$, and otherwise empty. Table S7 gives the finite-duration floors of Section 3.4: the smallest state from which n consecutive floor-class years are survivable under each rule, which is the n-step nominal boundary.

Table S6: Cod proportional-harvest family under the q10 floor: catch at the LRP, margin $C^* - C(\theta)$, b_∞ and regime.

ϕ	$C(\theta)$	margin	b_∞	regime
0.25	43.12	48.48	884.6	whole safe set
0.5	86.23	5.36	884.6	whole safe set
0.6	103.48	-11.89	1092.0	half-line
0.7	120.72	-29.13	1751.8	half-line
0.75	129.35	-37.75	∅	empty

Proportional rules $C = \phi\, g(S)$ above the LRP. Whole safe set iff $\phi \le \phi^*_{\mathrm{safe}} = 1 - |e|/g(\theta) = 0.5311$; non-empty iff $\phi \le 1 - |e|/g_{\max} = 0.7269$ (q10 class). Earlier editions printed the second threshold (0.727) for the first claim; the structural campaign corrected it. b_∞ is the smallest fixed point s_- of the worst-floor map (Theorem 2.3 of the main text); the archived structural campaign reports the finite-horizon proxies 1074.8 and 1568.4 kt for $\phi = 0.6$ and 0.7, which lie below s_- because the backward recursion of a proportional rule converges to s_- only geometrically with ratio $1/F'(s_-) \approx 0.95$ and 0.98.

Table S7: Cod: finite-duration floors.

policy	class	$n = 5$	10	15	20
BAU (5 kt)	worst	1450.0	1803.7	2062.3	2275.0
flat 60 kt	worst	1656.7	2188.8	2657.5	3177.4
S1	worst	1656.7	2188.8	2657.5	3177.4
cascade	worst	1656.7	2188.8	2657.5	3177.4
flat 120 kt	worst	1891.7	2671.8	3562.3	5311.8
BAU (5 kt)	q05	1298.7	1540.7	1697.8	1806.9
flat 60 kt	q05	1499.6	1893.2	2193.4	2456.2
S1	q05	1499.6	1893.2	2193.4	2456.2
cascade	q05	1499.6	1893.2	2193.4	2456.2
flat 120 kt	q05	1727.6	2329.0	2898.1	3619.7
BAU (5 kt)	q10	884.6	884.6	884.6	884.6
flat 60 kt	q10	884.6	884.6	884.6	884.6
S1	q10	884.6	884.6	884.6	884.6
cascade	q10	884.6	884.6	884.6	884.6
flat 120 kt	q10	979.9	1028.6	1053.9	1067.2

Lower boundary of the set of states from which n consecutive floor years can be absorbed and the nominal (zero-residual) map then holds the safe set. For $C < g(\theta)$ this coincides with the n-step kernel boundary b_n.

S6 Cod identification: profile in K, recent windows, alternative forms

Table S8 profiles the one-step least-squares fit over the carrying capacity K with r in closed form at each K, and reports the two profile cuts (one-parameter F cut 1.1955 and two-parameter cut 1.313 at the 95% level with $n = 24$), the implied sets of K and the ranges of $g(\theta)$, $F'(\theta)$ and C^* over them. Table S9 refits the map on post-moratorium windows of the 2016 series and on windows of the 2024 reassessment series with its own reference point (276 kt); the archived recent-window campaign stopped its optimiser at the box edge $K = 5000$ for 1995–2015, where the unconstrained minimiser is $K = 4316.6$ ($C^* = 165.7$ against 171.4 kt at the edge), and both are reported. Table S10 gives the Fox and Allee alternatives. The observation-error sensitivity of the archived identification campaign (a share λ of the residual variance attributed to observation error, which shrinks the floors) is reported in the notes to Table S12: the vacuous worst-floor class disappears for $\lambda \geq 0.2$, and at $\lambda = 0.5$ the maximal robust flat catch is 112.1 kt.

Table S8: Cod: profile of the one-step least-squares fit over K.

K (kt)	$r(K)$	SSE ratio	$g(\theta)$	$F'(\theta)$	C^*
1000	0.5094	1.7459	52.01	0.6082	-28.86
1275	0.4888	1.3009	132.39	0.8106	51.52
1500	0.4099	1.1941	148.77	0.9264	67.90
1775	0.3550	1.1342	157.55	1.0012	76.68
2000	0.3273	1.1068	161.47	1.0378	80.60
2500	0.2908	1.0730	166.20	1.0850	85.33
3000	0.2704	1.0551	168.66	1.1109	87.79
4000	0.2485	1.0366	171.18	1.1386	90.31
5000	0.2369	1.0271	172.46	1.1531	91.59
7000	0.2248	1.0174	173.75	1.1680	92.89
10000	0.2166	1.0109	174.62	1.1783	93.76
20000	0.2076	1.0039	175.55	1.1893	94.68
50000	0.2026	1.0000	176.06	1.1954	95.19

Closed-form $r(K)$ at each K; SSE ratio relative to the profile minimum (attained at the top of the grid). Cuts: one-parameter 1.1955, two-parameter 1.3130. Expansion $F'(\theta) > 1$ iff $K > 2\theta = 1769.2$ kt. Zero-budget point $(r, K) = (0.11107, 5000)$: SSE ratio 1.2793 to the profile minimum (1.2456 to the committed fit) – inside the two-parameter region, outside the one-parameter cut.

Table S9: Cod: refits on recent windows.

window	n	r	K	$g(\theta)$	e_{q10}	$e_{\min}$	C^*	C^*_{worst}	$F'(\theta)$	SD
NCAM 1995-2015 (exact profile minimiser)	20	0.2746	4316.6	193.1	-27.5	-32.6	165.7	160.6	1.162	18.9
NCAM 1995-2015 (K held at 5000)	20	0.2731	5000.0	198.9	-27.4	–	171.4	–	1.176	–
NCAM 1995-2007 (exact profile minimiser)	12	0.3901	5000.0	284.0	-10.0	-12.8	274.0	271.2	1.252	10.6
NCAM 1995-2007 (K held at 5000)	12	0.3901	5000.0	284.0	-10.0	–	274.0	–	1.252	–
xteNCAM 2005-2024 (archived; LRP 276 kt)	19	0.4439	472.0	50.9	-58.5	-88.8	-7.7	-37.9	0.925	40.4
xteNCAM 1995-2024 (archived; LRP 276 kt)	29	0.4310	472.0	49.4	-41.4	-88.2	8.0	-38.8	0.927	32.6

Refits on post-moratorium windows (source-year convention). For 1995–2015 the SSE profile is flat to within 0.001% between $K = 4317$ (exact minimiser) and the box edge where the archived optimiser stopped; both rows are shown. xteNCAM rows use the 2024 assessment series (Regular et al. 2025) with its own LRP of 276 kt; K sits at the lower bound of the box and the reference point is not self-viable under either floor.

Table S10: Cod: alternative surplus forms on the frozen window.

form	r	K	MSE (kt^2)	$g_{\max}$	$g(\theta)$	$F'(\theta)$	C^* (q10)
Schaefer (committed)	0.2369	5000.0	12772.2	296.1	172.46	1.1531	91.59
Allee (committed refit)	2.0	1671.66	7690.1	–	–	–	–
Fox (this campaign)	0.1044	5000.0	13873.1	192.0	159.92	1.0764	79.05

Archived forms campaign (source-year basis). The Allee refit is a boundary solution (r at its box edge of 2.0) and defines no admissible constants; the declared-s_0 Allee row and the data-preferred row give $C^* = 123.3$ and 115.2 kt and $C_{\text{vac}} = 802.8$ and 291.5 kt respectively (cadence campaign). Fox kernels under the worst floor: BAU $b_1 = 1047.7$, $b_5 = 1640.8$ kt, $T = \infty$ empty, as for Schaefer.

S7 Cod stochastic layer and bootstraps

Table S11 is the archived stochastic campaign ($N = 20\,000$ trajectories per cell): the probability that a trajectory started at S_0 under the registered map with resampled fit residuals stays at or above the LRP through T years, under i.i.d. resampling, block-of-four resampling and i.i.d. resampling with the 1992 residual removed, for the rule family and for a scan of constant catches. The ceiling $(23/24)^{20} = 0.4269$ (0.427 rounded) for never drawing the 1992 residual in twenty i.i.d. draws is the reason no rule exceeds 0.906. The independent re-run of the i.i.d. scheme in the core (seed 20261015) gives 0.9031, 0.8353 and 0.6466 for BAU, 60 kt and 120 kt at $T = 20$ from the

LRP. Table S12 gives the fixed-K and joint trajectory bootstraps of C^* (archived, $B = 2000$) and the independent fixed-K re-run (seed 20261015), under a frozen floor and a re-estimated floor.

Table S11: Cod stochastic layer (archived campaign, $N = 20\,000$ per cell): probability of staying at or above the LRP through T years from S_0.

scheme	S_0	T	BAU	zero	60 kt	120 kt	S1	cascade
iid	884.6	5	0.9002	0.9023	0.8324	0.6624	0.8324	0.8324
iid	884.6	10	0.9013	0.9034	0.8354	0.6464	0.8354	0.8354
iid	884.6	20	0.9031	0.9056	0.8353	0.6466	0.8353	0.8353
iid	861.9	5	0.8863	0.8994	0.8266	0.6072	0.8266	0.8266
iid	861.9	10	0.8843	0.8969	0.8247	0.5803	0.8247	0.8247
iid	861.9	20	0.8832	0.8958	0.8246	0.5805	0.8246	0.8246
iid	1000	5	0.9157	0.9157	0.9008	0.8217	0.9008	0.9008
iid	1000	10	0.9156	0.9157	0.8998	0.8084	0.8998	0.8998
iid	1000	20	0.9141	0.9141	0.8994	0.8033	0.8994	0.8994
iid	1500	5	1.0000	1.0000	0.9999	0.9992	0.9999	0.9999
iid	1500	10	1.0000	1.0000	1.0000	0.9986	1.0000	1.0000
iid	1500	20	1.0000	1.0000	0.9999	0.9990	0.9999	0.9999
block4	884.6	5	0.8517	0.8517	0.8090	0.7483	0.8090	0.8090
block4	884.6	10	0.8537	0.8537	0.8091	0.6606	0.8091	0.8091
block4	884.6	20	0.8515	0.8515	0.8065	0.6499	0.8065	0.8065
block4	861.9	5	0.8532	0.8532	0.8101	0.6680	0.8101	0.8101
block4	861.9	10	0.8514	0.8515	0.8100	0.6155	0.8100	0.8100
block4	861.9	20	0.8425	0.8428	0.7962	0.5937	0.7962	0.7962
block4	1000	5	0.8589	0.8589	0.8116	0.8116	0.8116	0.8116
block4	1000	10	0.8575	0.8599	0.8128	0.7612	0.8128	0.8128
block4	1000	20	0.8542	0.8562	0.8081	0.7519	0.8081	0.8081
block4	1500	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
block4	1500	10	1.0000	1.0000	1.0000	0.9791	1.0000	1.0000
block4	1500	20	1.0000	1.0000	1.0000	0.9792	1.0000	1.0000
iid-no1992	884.6	5	0.9522	0.9522	0.9167	0.7772	0.9167	0.9167
iid-no1992	884.6	10	0.9505	0.9506	0.9134	0.7624	0.9134	0.9134
iid-no1992	884.6	20	0.9537	0.9538	0.9172	0.7694	0.9172	0.9172
iid-no1992	861.9	5	0.9393	0.9459	0.9104	0.7158	0.9104	0.9104
iid-no1992	861.9	10	0.9420	0.9499	0.9096	0.7118	0.9096	0.9096
iid-no1992	861.9	20	0.9413	0.9489	0.9103	0.7033	0.9103	0.9103
iid-no1992	1000	5	0.9539	0.9539	0.9483	0.9079	0.9483	0.9483
iid-no1992	1000	10	0.9546	0.9546	0.9483	0.9051	0.9483	0.9483
iid-no1992	1000	20	0.9533	0.9533	0.9469	0.9020	0.9469	0.9469
iid-no1992	1500	5	1.0000	1.0000	1.0000	0.9998	1.0000	1.0000
iid-no1992	1500	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
iid-no1992	1500	20	1.0000	1.0000	1.0000	0.9997	1.0000	1.0000

Constructive-bound scan (flat catch C from the LRP, 20 years): survival crosses 0.9 at 13.5 kt (i.i.d.), never (blocks), 78.9 kt (no 1992); 0.8 at 81.2 / 72.3 / 105.2 kt; at 92.5 kt survival is 0.742 / 0.727 / 0.844.

Table S12: Cod: residual bootstraps of the fit and of the constructive bound.

design	r median	r 90%	C^* median	C^* 90%	$C^* \leq 0$
fixed-K (archived, $B = 2000$)	0.2192	[0.0159, 0.2774]	78.70	[-69.29, 121.07]	235/2000
fixed-K (independent seed, frozen floor)	0.2193	[0.0254, 0.2764]	78.81	[-62.37, 120.34]	12.5%
fixed-K (independent seed, floor re-estimated)	0.2193	–	115.25	[-66.12, 171.55]	9.8%
joint (r, K) (archived, $B = 2000$)	0.2611	[0.0924, 0.4997]	73.72	[-89.38, 125.66]	387/2000
joint, $K \geq 2\theta$ (1457 refits)	–	–	88.14	[-5.59, 130.56]	83/1457

Joint design: median K 4191 kt, 41.7% of refits pinned at 5000 kt, 72.9% expansive. Observation-error sensitivity (archived identification campaign; a share λ of residual variance attributed to measurement error and removed before the floor is taken): $\lambda = 0.0$: C^* =91.6; $\lambda = 0.1$: C^* =95.2; $\lambda = 0.2$: C^* =99.0; $\lambda = 0.3$: C^* =103.0; $\lambda = 0.4$: C^* =107.4; $\lambda = 0.5$: C^* =112.1 kt.

S8 Edwards nominal and certified kernels

Table S13 gives, for both thresholds, every rule and every floor, the nominal kernel boundaries at the printed horizons, the first empty horizon, and the Lipschitz-certified boundaries (threshold inflated by $r_T = \varepsilon(1 - a^T)/(1 - a)$ with the declared $\varepsilon = 15.41$ ft). The 660-ft kernels are empty in finite time for every rule because the zero-pumping attractor lies below 660 ft under every floor (Theorem 2.10); the BAU, Stage-I and cascade 660-ft kernels coincide because all three pump $\bar{P}$ at and above 660 ft.

Table S13: Edwards Aquifer: nominal kernel boundaries (ft) by threshold, policy, floor and horizon, first empty horizon, and Lipschitz-certified boundaries at $T = 1$–5 ($\varepsilon = 15.41$ ft).

thr.	policy	floor	T=1	2	3	5	8	10	empty	cert. 1	2	3	4	5
618	BAU ($\bar{P}$)	worst	618.8	619.8	621.2	625.6	639.5	658.4	13	639.4	668.2	706.7	∅	∅
618	BAU ($\bar{P}$)	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	636.2	660.5	693.2	∅	∅
618	BAU ($\bar{P}$)	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	635.8	659.7	691.8	∅	∅
618	flat 90%	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	638.4	665.6	702.2	∅	∅
618	flat 90%	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	635.1	658.0	688.7	∅	∅
618	flat 90%	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	634.8	657.2	687.3	∅	∅
618	flat 80%	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	637.3	663.1	697.8	∅	∅
618	flat 80%	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	634.0	655.5	684.3	∅	∅
618	flat 80%	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.7	654.7	682.9	∅	∅
618	flat 70%	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	636.2	660.6	693.3	∅	∅
618	flat 70%	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	653.0	679.8	∅	∅
618	flat 70%	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	652.2	678.4	∅	∅
618	flat 60%	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	635.1	658.1	688.9	∅	∅
618	flat 60%	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	650.5	675.4	708.7	∅
618	flat 60%	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	649.7	674.0	706.6	∅
618	flat 50%	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	634.1	655.6	684.4	∅	∅
618	flat 50%	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	647.9	670.9	701.7	∅
618	flat 50%	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	647.2	669.5	699.5	∅
618	zero pumping	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	644.9	662.2	687.9	∅
618	zero pumping	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	644.9	653.5	666.5	690.4
618	zero pumping	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	644.9	653.5	664.3	687.2
618	Stage I	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	637.3	665.3	706.7	∅	∅
618	Stage I	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	634.0	655.5	693.2	∅	∅
618	Stage I	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.7	654.7	691.8	∅	∅
618	CPM cascade	worst	618.0	618.0	618.0	618.0	618.0	618.0	never	635.7	665.3	706.7	∅	∅
618	CPM cascade	q05	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	654.0	693.2	∅	∅
618	CPM cascade	q10	618.0	618.0	618.0	618.0	618.0	618.0	never	633.4	653.3	691.8	∅	∅
660	BAU ($\bar{P}$)	worst	675.1	695.3	∅	∅	∅	∅	3	695.7	∅	∅	∅	∅
660	BAU ($\bar{P}$)	q05	671.8	687.6	708.8	∅	∅	∅	4	692.5	∅	∅	∅	∅
660	BAU ($\bar{P}$)	q10	671.5	686.8	707.5	∅	∅	∅	4	692.1	∅	∅	∅	∅
660	flat 90%	worst	674.0	692.8	∅	∅	∅	∅	3	694.6	∅	∅	∅	∅
660	flat 90%	q05	670.7	685.1	704.4	∅	∅	∅	4	691.4	∅	∅	∅	∅
660	flat 90%	q10	670.4	684.3	703.0	∅	∅	∅	4	691.1	∅	∅	∅	∅
660	flat 80%	worst	672.9	690.2	∅	∅	∅	∅	3	693.6	∅	∅	∅	∅
660	flat 80%	q05	669.7	682.6	699.9	∅	∅	∅	4	690.3	∅	∅	∅	∅
660	flat 80%	q10	669.3	681.8	698.6	∅	∅	∅	4	690.0	∅	∅	∅	∅
660	flat 70%	worst	671.8	687.7	709.0	∅	∅	∅	4	692.5	∅	∅	∅	∅
660	flat 70%	q05	668.6	680.1	695.5	∅	∅	∅	4	689.2	∅	∅	∅	∅
660	flat 70%	q10	668.2	679.3	694.1	∅	∅	∅	4	688.9	∅	∅	∅	∅
660	flat 60%	worst	670.8	685.2	704.5	∅	∅	∅	4	691.4	∅	∅	∅	∅
660	flat 60%	q05	667.5	677.6	691.0	∅	∅	∅	5	688.2	∅	∅	∅	∅
660	flat 60%	q10	667.2	676.8	689.7	∅	∅	∅	5	687.8	∅	∅	∅	∅
660	flat 50%	worst	669.7	682.7	700.1	∅	∅	∅	4	690.3	∅	∅	∅	∅
660	flat 50%	q05	666.4	675.0	686.6	∅	∅	∅	5	687.1	∅	∅	∅	∅
660	flat 50%	q10	666.1	674.3	685.2	∅	∅	∅	5	686.7	∅	∅	∅	∅
660	zero pumping	worst	664.3	670.1	677.8	702.2	∅	∅	6	685.0	∅	∅	∅	∅
660	zero pumping	q05	661.1	662.5	664.3	670.3	689.1	∅	10	681.7	∅	∅	∅	∅
660	zero pumping	q10	660.7	661.7	663.0	667.0	679.8	697.3	11	681.4	∅	∅	∅	∅
660	Stage I	worst	675.1	695.3	∅	∅	∅	∅	3	695.7	∅	∅	∅	∅
660	Stage I	q05	671.8	687.6	708.8	∅	∅	∅	4	692.5	∅	∅	∅	∅
660	Stage I	q10	671.5	686.8	707.5	∅	∅	∅	4	692.1	∅	∅	∅	∅
660	CPM cascade	worst	675.1	695.3	∅	∅	∅	∅	3	695.7	∅	∅	∅	∅
660	CPM cascade	q05	671.8	687.6	708.8	∅	∅	∅	4	692.5	∅	∅	∅	∅
660	CPM cascade	q10	671.5	686.8	707.5	∅	∅	∅	4	692.1	∅	∅	∅	∅

S9 Edwards bounded-defect audit

Table S14 gives the exact bounded-defect kernel boundaries at 618 ft under the worst floor (Theorem 2.7) for the declared defect (15.41 ft, fit window) and the out-of-sample defect (21.81 ft), and the supremum uniform defect that each horizon can carry. For the constant-pumping rules (affine closed-loop maps) the exact boundaries coincide with the Lipschitz-certified boundaries of Section S8 wherever the recursion binds (Proposition 2.8(ii)). The departures are listed in the table: for zero pumping at $T = 1, 2$ the certificate's inflation of the initial state binds instead of the recursion; for the Stage-I rule ($T = 3$) and the cascade ($T = 2, 3$) the closed-loop map is piecewise affine and the exact kernel is larger by 2–7 ft. No departure changes a horizon. At $T = 3$ the supremum defects are 16.01 ft for BAU, 16.71 ft for Stage I, 17.23 ft for the cascade, 17.62 ft for flat 80% and 24.04 ft for zero pumping; at $T = 4$ zero pumping alone retains a nonempty kernel, up to 17.93 ft. The 21.81-ft out-of-sample defect leaves only zero pumping certified through $T = 3$, and even that rule fails at $T = 4$.

Table S14: Edwards: exact bounded-defect kernel boundaries at 618 ft (worst floor) for the fit-window and out-of-sample defects.

policy	ε (ft)	T=1	2	3	4	5	first empty
BAU ($\bar{P}$)	15.41 (fit)	639.4	668.2	706.7	$\emptyset$	$\emptyset$	4
BAU ($\bar{P}$)	21.81 (OOS)	648.0	688.2	$\emptyset$	$\emptyset$	$\emptyset$	3
flat 90%	15.41 (fit)	638.4	665.6	702.2	$\emptyset$	$\emptyset$	4
flat 90%	21.81 (OOS)	646.9	685.7	$\emptyset$	$\emptyset$	$\emptyset$	3
flat 80%	15.41 (fit)	637.3	663.1	697.8	$\emptyset$	$\emptyset$	4
flat 80%	21.81 (OOS)	645.9	683.2	$\emptyset$	$\emptyset$	$\emptyset$	3
Stage I (−20% below 660 ft)	15.41 (fit)	637.3	665.3	702.8	$\emptyset$	$\emptyset$	4
Stage I (−20% below 660 ft)	21.81 (OOS)	645.9	685.3	$\emptyset$	$\emptyset$	$\emptyset$	3
CPM cascade	15.41 (fit)	635.7	663.1	699.9	$\emptyset$	$\emptyset$	4
CPM cascade	21.81 (OOS)	644.8	683.9	$\emptyset$	$\emptyset$	$\emptyset$	3
zero pumping	15.41 (fit)	628.7	643.0	662.2	687.9	$\emptyset$	5
zero pumping	21.81 (OOS)	637.3	663.1	697.7	$\emptyset$	$\emptyset$	4

Fit-window defect $\varepsilon = 15.41$ ft; out-of-sample defect 21.81 ft; threshold 618 ft, ceiling 710 ft, recharge pinned at the 1956 floor. For positive flat-cap rules the fit-window rows coincide with the Lipschitz-certified boundaries of Table S13 at every reported horizon; zero pumping coincides from $T = 3$ onward (Proposition 2.8(ii)). Departures (exact vs certified, ft): Stage I $T = 3$: 702.8 vs 706.7; cascade $T = 2$: 663.1 vs 665.3; cascade $T = 3$: 699.9 vs 706.7; zero pumping $T = 1$: 628.7 vs 633.4; zero pumping $T = 2$: 643.0 vs 644.9. For zero pumping the certificate's inflated initial-state constraint binds at $T = 1, 2$; for the staged rules the closed-loop map is piecewise affine. These boundary departures do not change a horizon.

S10 Edwards replays, closed loop, finite floors and floor-class supply

Table S15 collects the 1951–1956 replays from the observed 1950 head, the closed-loop end and minimum heads for 1934–1990 and 1990–2023, the finite-duration floors (the smallest head from which n consecutive 1956 years are absorbed) and the floor-class supply (57 years of recharge pinned at the 1956 floor from the 1934 head). The observed 1956 annual mean is 623.15 ft (daily minimum 612.51 ft); the model with the actual 1950s pumping reaches 631.2 ft, a bias of 8.1 ft in the unsafe direction and within the declared defect.

Table S15: Edwards: replays, closed-loop end heads and floor-class supply.

policy	1950s min. head	closed-loop min. 1934–90	end 1990	end 2023	floor-class supply
BAU ($\bar{P}$)	629.65	626.67	677.65	639.20	282.16
flat 90%	632.27	629.82	680.81	642.35	–
flat 80%	634.88	632.98	683.97	645.50	225.73
flat 70%	637.50	–	–	–	–
flat 60%	640.11	639.29	689.64	651.76	169.29
flat 50%	642.73	–	–	–	–
zero pumping	655.81	658.22	700.73	670.55	–
Stage I (−20% below 660 ft)	634.88	632.88	678.03	643.32	226.72
CPM cascade	637.06	636.20	678.03	644.26	174.49

1950s replay: model from the observed 1950 head with actual 1951–1956 recharge and the rule's pumping (minimum head reached; observed 1956 minimum 623.15 ft). Closed loop: observed recharge, rule pumping, from the 1934 head (fit) or 1990 head (OOS); observed end heads 645.8 (1990) and 635.7 (2023) ft. Floor-class supply: 57 closed-loop years with recharge pinned at the worst floor from the 1934 head. Open-loop fit diagnostic with actual pumping: maximum absolute error 8.1 ft (1956: model 631.2, observed 623.2). Stage occupancy of observed heads below 660/650/640/630 ft: fit window [20, 12, 5, 1] of 57; out of sample [11, 5, 2, 0] of 33. Finite-duration floors at 618 ft (worst floor then mean recharge 636.7): BAU $n = 5$: 625.6 ft, $n = 10$: 658.4 ft, $n = 15$: empty; every cut policy: whole safe set.

S11 Edwards residual bootstrap of the attractors

Table S16 gives the archived residual bootstrap of the 1934–1990 fit ($B = 5000$, seed 20260905) and the independent re-run in the core (seed 20261015): the bootstrap mean, standard deviation and 90% interval of each rule's worst-floor attractor and the probability that it clears 618 ft. No replicate is non-contractive, $\gamma < 0$ in every replicate, and the flat-90% minus BAU gap is positive in every replicate (minimum 0.5 ft, 90% interval $[1.9, 5.1]$ ft). The ratios $\beta/(1-a)$ and $\gamma/(1-a)$ have bootstrap coefficients of variation of 0.25 and 0.30; the ordering of the rules depends only on the sign of γ, the level of the attractors on the ratios.

Table S16: Edwards: residual bootstrap of the worst-floor attractors.

policy	H^*	mean	SD	[q05, q50, q95]	$P(H^* \geq 618)$
flat-90%	618.88	617.00	12.46	[594.49, 619.02, 633.04]	0.5336
flat-80%	622.04	620.27	11.87	[598.69, 622.09, 635.69]	0.6500
flat-70%	625.20	623.55	11.35	[603.09, 625.33, 638.27]	0.7486
flat-60%	628.36	626.83	10.89	[607.27, 628.41, 641.23]	0.8290
flat-50%	631.52	630.11	10.51	[611.25, 631.55, 644.18]	0.8864
flat-0%	647.32	646.49	10.03	[628.99, 647.33, 661.16]	0.9894
BAU	615.72	613.72	13.09	[590.01, 615.82, 630.55]	0.4148
S1	622.04	620.27	11.87	[598.69, 622.09, 635.69]	0.6500
CPM	628.36	626.19	10.32	[607.27, 628.41, 639.67]	0.8290

Archived audit layer, 5000 replications, seed 20260905, no non-contractive replicate. Independent re-run (seed 20261015): $P(H^* \geq 618)$ BAU 0.403, flat 90% 0.523, flat 80% 0.644, flat 60% 0.824, zero 0.989; the flat-90% minus BAU attractor gap has 90% interval [1.9, 5.1] ft (point value 3.16 ft).

S12 Verification summary

Table S17 groups the 926 agreement checks of the core against the archived artefacts of the registered protocols (campaign CSV and JSON files shipped in the source folder) by artefact, with the tolerance and the largest observed discrepancy in each group. The archived campaign scripts were re-executed on the locked panels during the preparation of this edition; the 26 result files they produce are byte-identical to the archived ones. The shipped verifier `verify_paper09_v44.py` re-runs the core and checks the manuscript's statements against it (Section 7).

Table S17: Agreement of the independent core with the registered artefacts, by group.

group	checks	passed
cod C*	1	1
cod K	1	1
cod a_max	1	1
cod e_min	1	1
cod e_q05	1	1
cod e_q10	1	1
cod finite	30	30
cod fixed-K	2	2
cod kernel	144	144
cod oos	1	1
cod r	1	1
cod stochastic	6	6
cod train	1	1
edw 1950s	1	1
edw P_bar	1	1
edw a	1	1
edw alpha	1	1
edw attractor	27	27
edw beta	1	1
edw bootstrap	5	5
edw certified	324	324
edw closed-loop	20	20
edw delta	1	1
edw floor	3	3
edw gamma	1	1
edw kernel	324	324
edw oos	1	1
edw replay	9	9
edw rho*	1	1
edw supply	9	9
edw train	1	1
recent NCAM	4	4
total	926	926

Tolerances: closed-form constants 10^{-3}–10^{-6}; kernel boundaries 0.05 kt / 0.01 ft (0.2 kt at $T = 20$); Monte-Carlo and bootstrap quantities re-run with an independent seed, tolerances 0.015–0.02 in probability and 3 kt in the bootstrap median.

S13 The two frozen protocols

The two intervention-selection protocols are reproduced below exactly as archived in the repository (`wave_e_cod/protocol_intervention.md` and `wave_e_edwards/protocol_intervention.md`), transliterated to ASCII and re-wrapped for the page; both were frozen on 2026-08-26 before any kernel, boundary, replay or retention score was computed. The main text follows them in every declared quantity; the exact bounded-defect kernels and the Lipschitz-versus-exact identity are additions that leave the declared classes unchanged.

S13.1 Northern cod (NAFO 2J3KL)

```
# Wave E intervention-selection protocol -- Northern cod (NAFO 2J3KL), Omega_2016

**Frozen 2026-08-26, before any kernel, boundary, replay, or retention score was computed.**
This is the cod-side analogue of ‘wave_e_edwards/protocol_intervention.md‘ (G1 Track 2,
G1a). It executes Sec.15’s intervention-selection leg on the **governed surplus-production
```

object** of ‘wave_e_cod‘ -- the ladder’s own model class, not NCAM. The ladder’s forecast verdicts (persistence wins; no module retained) are inputs, not outputs, of this protocol and are not re-litigated.

1. Object (declared)

Field	Contents	Type
System	Northern cod 2J3KL as represented by NCAM M-shift SSB, 1983-2015 (the locked Omega_2016 series; not pooled with xteNCAM)	D
Map	Discrete surplus with catch: ‘S_{t+1} = [S_t + r*S_t(1-S_t/K) - C_t + e_t]+‘ (the ladder’s M2 class, Allee off: s = 0)	M
Fit window	1983-2007, one-step least squares on Schijns annual catch (the ladder’s own ‘fit_params‘, bounds r in (0.001, 2], K in (max train, 5000])	E
OOS audit window	2008-2015 (recovery era): residuals audited, no refit	E
Safe set	K* = LRP = 884.6 kt (the 1983-1989 mean, the declared [N] threshold of Omega_2016). Omega_2016 declares **no second threshold** -- there is no cod analogue of the Edwards phys/inst pair, and the 2023 40% B_MSY LRP belongs to Omega_xte and is not pooled	N
State domain	[1e-3, 10000] kt (the ladder’s clip floor; upper bound declared at twice the fitted K)	M
Governance family	Sec.4 below	E/[N]

Declared defects carried by the object. (i) K is expected to pin at its optimization upper bound (the series never approaches carrying capacity) -- all upper edges inherit it; the LRP-boundary results depend chiefly on r, which is identified. (ii) The residual conflates productivity shocks and Schaefer-class model error -- there is no independent input channel (unlike Edwards’ recharge), so the disturbance classes and the defect declaration below are the **same measured quantity in two roles**; disclosed, not repaired. (iii) One pool, no age structure, no migration (A014-L list).

2. Defect declaration (Cor2 triple, first element)

eps = max |residual| over the **fit window** 1983-2007, in kt yr-1, together with the residual SD and the lower-tail quantiles (the UC classes below). The OOS audit (2008-2015) is reported alongside; no refit is permitted whatever it shows.

3. Uncertainty classes (persistent productivity shocks)

Mirroring the Edwards recharge floors (perpetual 1956 / q05 / q10 of training recharge), the disturbance classes are **persistent additive productivity floors** drawn from the fit-window residual distribution:

Class	Floor e (kt yr-1)	Reading
UC-min	min(fit residuals)	perpetual worst observed one-step productivity shock
UC-q05	5th percentile of fit residuals	perpetual 5th-percentile shock
UC-q10	10th percentile of fit residuals	perpetual 10th-percentile shock

The worst-case closed loop is ‘F(S) = [S + r*S(1-S/K) - C(S) + e]+‘ with e the class floor (a constant -- the persistent reading, deliberately harsher than i.i.d.).

4. Governance-operator family

ID	Rule	Type
BAU	C ? 5 kt (moratorium-level inshore removals; the declared implementable U post-1992)	E
flat_100 ... flat_0	C ? rho*240 kt, rho in {1.0, 0.75, 0.5, 0.25, 0.0} (240 kt = pre-1992 directed-fishery level)	E
S1	C = 60 kt if S >= LRP, else 0 (DFO 2009 PA critical-zone rule at a modest declared cap)	E/[N]
cpm	cascade: C = 60 kt if S >= LRP; 30 kt if S in [0.75*LRP, LRP); 5 kt if S in [0.5*LRP, 0.75*LRP); 0 below 0.5*LRP	[N] (declared scenario)

```
Stage thresholds below the LRP are declared scenarios, not verified institutions
(the Edwards analogue: stages II-IV declared [N]).

## 5. Kernels

Robust T-step viability kernel of the safe set [K*, S_HI] under the persistent floor:
K_0 = [K*, 10000]; K_{n+1} = {S in K_n : F(S) in K_n} with F the worst-case closed loop
(policies act on the current state; the catch jump thresholds split the domain into
pieces, on each of which F is a concave quadratic -- preimages are computed by interval
arithmetic on the quadratic roots). Horizons T in {1, 2, 3, 5, 8, 10, 15, 20, inf}; the
infinite-horizon kernel is the fixpoint with a one-step stability re-check.

## 6. Certified layer (Cor2/Cor5 erosion conversion)

r_T = eps*(1 - a^T)/(1 - a) if the closed loop contracts with rate a < 1 on the safe
domain; **otherwise** the expansive form r_T = eps*(a_max^T - 1)/(a_max - 1) with
a_max = sup|F'| over the safe domain. The form that applies is computed from the
fitted map and **recorded** -- if the map is expansive at the safe-set boundary (as the
Schaefer map is below K/2), the contraction form is inapplicable and the certified
kernel is the nominal kernel of K* + r_T with the expansive r_T. No certified claim at
any horizon where K* + r_T exceeds the nominal kernel's reach.

## 7. Replays

1. **Supply replay** (no dynamics): mean allowed catch C(S_t^obs) over the fit-window
   states 1983-2006, per policy; cut-active fraction reported.
2. **Stress replay** (the 1990s analogue of the Edwards 1950s replay): closed-loop
   model replay from the observed 1990 SSB with the **observed** 1991-1995 residuals,
   per policy; whether the path stays >= LRP reported. (The ladder already established
   that catch cannot produce the 1992-94 pulse; this replay records what governance
   does against the *observed* shock class.)
3. **T=5 classification**: which observed SSB values lie outside each policy's T=5
   nominal kernel, per UC class.

## 8. Retention rule (frozen; mirrors the Edwards rule)

A non-BAU policy is **retained** iff:

- (a) its robust kernel is at least as protective as BAU's at **every** (UC, T)
  reading -- compared on the kernel lower boundary, empty = worst; and
- (b) it improves on BAU somewhere (strictly lower boundary at some reading); and
- (c) at some reading where it improves, its supply-replay mean catch exceeds that of
  every at-least-as-protective flat cap (the "matched protection" clause).

No forecast module is promoted or demoted. The oracle, the survey-start variant, and
capelin modules play no role. No Omega_xte row is produced.

## 9. Outputs

`results/intervention_results.json` (fit, UC, kernels, steady states, maximal robust
flat catch, supply, stress replay, classification, retention at nominal and certified
levels, certified horizons), `results/intervention_boundaries.csv` (per policy x UC x T
kernel lower boundaries, nominal and certified), admission row
`admission/R04_Cor2_cod_kernel.md`, manuscript `manuscript/wave_E_cod_intervention.md`.
Deterministic; no randomness anywhere.
```

S13.2 Edwards Aquifer (J-17)

```
# Wave E -- Edwards San Antonio Pool. Intervention-selection protocol

**Locked 2026-08-26, before the intervention scores were generated.**
Companion to `protocol.md` (the forecast-ladder protocol, locked 2026-08-25).
```

This is the Sec.15 **intervention-selection leg** -- never before exercised on a
real system in this programme. It shares the locked data object of
‘protocol.md‘ and adds a governance-operator comparison with viability
kernels. It does **not** reopen any forecast-ladder retention decision.

Protocol sentence (frozen)

Intervention retention is decided solely by robust viability kernels and
replayed supply under the declared uncertainty classes, against the
business-as-usual baseline. No forecast module is promoted or demoted by
this leg; the fibre and the oracle stay excluded; no two-pool claim is made.

Object (unchanged from ‘protocol.md‘)

Field	Contents	Type
\(S\)	Edwards Aquifer, San Antonio Pool, as indexed by well J-17	D
\(z_t\)	Calendar-year mean of daily-high J-17 elevation, ft AMSL	D
\(B\)	San Antonio Pool management region; calendar years 1934-2023	D
Data	‘data/annual_panel.csv‘ only (locked 20-column panel); no new data	D
\(K^*_{\mathrm{phys}}\)	618 ft (Comal cessation proximity; declared, not certified)	N / E
\(K^*_{\mathrm{inst}}\)	660 ft (post-2007 Stage I trigger; not applied pre-2007)	N
Not \(z\)	Comal, GRACE, J-27, San Marcos, MODFLOW/GWSIM heads	D

Dynamics (the scored ladder’s M2 class, one pool, affine)

\[
\Delta H_t = \alpha + \beta R_t + \gamma P_t + \delta H_{t-1},
\qquad a = 1+\delta,
\]
fitted by ordinary least squares on the training window 1934-1990
(transitions with both endpoints \(\le\) 1990). Out-of-sample years
1991-2023 are used **only** for the defect audit; no refitting.

Model domain (declared compact set, the ladder’s clip bounds):
\(H \in [610, 710]\) ft. Upward exits above 710 ft under extreme recharge
are out-of-domain for the *model* and are not violations of the
lower-threshold constraint; this caveat is carried in the admission record.

Information pattern (causal, matches the H0 record’s map (5))

At the end of year \(t\) the manager observes \((H_t, R_t, P_t)\) and sets
next year’s pumping \(P_{t+1} = \pi(H_t)\) (annual-mean approximation to the
intra-year CPM adjustment). Next year’s recharge \(R_{t+1}\) is unknown at
decision time and is treated adversarially.

Uncertainty classes (declared, robust, persistent recharge floors)

Computed on the training window only:

Class	Declaration	Value (103 acre-ft yr-1)
UC-min	\(R \ge \min_{\text{train}} R\) (1956 drought-of-record year, held forever)	43.7
UC-q05	\(R \ge\) 5th percentile of train \(R\)	166.5
UC-q10	\(R \ge\) 10th percentile of train \(R\)	179.1

For the lower-threshold constraint the adversarial realisation is the floor
(\(\beta > 0\) verified on the fit). These floors are **persistent**
(forever), which is harsher than any recorded drought; the classes are
declared for certification geometry, not as recharge forecasts.

Governance operators (declared family)

\(\bar P \) = training-mean pumping (1934-1990). Stage thresholds and the
Stage I reduction are from ‘data/SOURCES.md‘; Stage II-IV reductions are
declared [N] scenarios, not verified in-repo.

ID	Rule	Note
BAU	\(P \equiv \bar P\)	baseline (the persistence analogue)
flat-\(\rho\)	\(P \equiv \rho\,\bar P\), \(\rho \in \{0.9, 0.8, 0.7, 0.6, 0.5, 0.0\}\)	flat caps
S1	\(P(H) = 0.8\,\bar P\) if \(H < 660\), else \(\bar P\)	post-2007 Stage I, 20% (verified)
cpm	cumulative cuts 20/30/35/40% at \(H<660/650/640/630\)	cascade, stages II-IV [N]

Scores (frozen)

1. **Nominal robust kernel boundary** \(b_T(\pi)\) (ft; lower = more protective): the infimum of the \(T\)-year robustly viable set \(\mathrm{Viab}_T(\pi, K^*)\) under each UC, computed by backward recursion \(K_0 = [K^*, 710]\), \(K_{n+1} = \{H \ge K^* : aH + \alpha + \beta R_{\mathrm{lo}} + \gamma\pi(H) \in K_n\}\), \(T \in \{1,2,3,5,8,10,15,20,\infty\}\). Empty set recorded as ‘null‘.
2. **Certified boundary** after the R04.Cor2 / R03.Cor5 erosion (discrete-contraction form): \(b^{\mathrm{cert}}_T = b_T(\pi; K^* + r_T)\) with \(r_T = \varepsilon (1-a^T)/(1-a)\), \(\varepsilon = \max_{\text{train}} |\text{residual}|\) (uniform defect declaration). Audits, not used for certification: the out-of-sample max residual (defect-adequacy check) and the training residual SD (non-uniform reading).
3. **Supply**: mean annual pumping under (i) the deterministic replay of the historical train recharge sequence 1934-1990 from \(H_{1934}\) and (ii) the OOS sequence 1991-2023 from \(H_{1990}\) (audit only).
4. **Stress replays**: the 1950-1956 recharge sequence from \(H_{1950}\) under each policy -- minimum head reached, and whether it stays \(\ge 618\) ft; plus classification of the actual 1956 and 2011-2014 annual states against the nominal and certified kernels.

Retention rule (frozen; mirrors the ladder’s persistence benchmark)

BAU is the baseline. A governance module is **retained** iff

(a) it is at least as protective as BAU on the nominal kernel boundary at **every** declared horizon under **every** declared UC and both declared \(K^*\) readings (no horizon/class/threshold where it is strictly worse), **and**

(b) it supplies strictly more water (train replay) than the most protective flat cap with the same nominal protection profile.

Otherwise the flat cap (or BAU) suffices -- "complexity not earned".
A module that improves only horizons where BAU is already certified earns no certified-increment claim. A policy whose certified kernel is empty at every horizon earns no certified claim at all (a nominal-only row).

Prohibitions (frozen)

- No forecast-module promotion (M2 stays rejected, M1 thin; oracle and Comal fibre stay excluded from every score here).
- No two-pool, karst, or solute claim; no pooling with Northern cod or any phosphorus catchment.
- \(K^*_{\mathrm{inst}}\) is not applied to pre-2007 history.
- Nothing here is ‘EXACT_SPECIALIZATION‘ of A005; every kernel statement carries the Cor2 triple (defect, horizon, erosion) and the

`'APPROXIMATION'` mapping type.

S14 Symbols and conventions

symbol	meaning
x_t, u_t, w_t	state, control (catch C_t or pumping P_t), driver (residual e_t or recharge R_t)
$F(x, u, w)$	registered one-step map; $\underline{F}(x, u) = F(x, u, \underline{w})$ its worst-floor map
$W_0 = [\theta, \bar{x}]$	safe set: states at or above the threshold θ and at or below the domain ceiling $\bar{x}$ (10 000 kt; 710 ft)
$W_T(\pi)$, $W_\infty(\pi)$	closed-loop T-step and infinite-horizon robust viability kernels of rule π (Definition 2.1)
b_T	lower boundary of a half-line kernel $W_T = [b_T, \bar{x}]$
s_-	smallest fixed point of the worst-floor map in the safe set
T_{empty}	first horizon at which a kernel is empty ("first empty"); the last non-empty horizon is $T_{\text{empty}} - 1$
ε	declared uniform defect: largest absolute fit residual (328.97 kt; 15.41 ft); out-of-sample defect 46.7 kt; 21.81 ft
r_T, r_∞	Lipschitz accumulation $\varepsilon \sum_{k<T} L^k$ with $L = F'(\theta)$ (cod) or $L = a$ (Edwards); $r_\infty = \varepsilon/(1-a)$
W_T^ε	exact bounded-defect kernel (Theorem 2.7)
T_{cert}	certified horizon: last T with $W_T^\varepsilon \neq \emptyset$ (exact) or with the Lipschitz certificate non-empty
S_t, C_t, $g(S) = rS(1 - S/K)$	cod spawning stock biomass (kt), catch (kt), Schaefer surplus; $r = 0.2369$, $K = 5000$ (pinned)
$\theta = 884.58$ kt	limit reference point (1983–1989 mean SSB)
$\lvert e \rvert$	floor-class magnitude: 328.97 (worst), 287.36 (q05), 80.87 kt (q10)
$C^* = g(\theta) - \lvert e \rvert$	maximal constant catch whose kernel is the whole safe set
$C_{\text{vac}} = g_{\max} - \lvert e \rvert$	maximal constant catch with a non-empty infinite-horizon kernel; $g_{\max} = rK/4$
ϕ^*_{safe}, ϕ^*_{nonempty}	proportional-rule thresholds $1 - \lvert e \rvert / g(\theta)$ and $1 - \lvert e \rvert / g_{\max}$
BAU, flat c, S1, cascade, graded	5 kt; constant c kt (60/120/180/240 = 25/50/75/100% of 240); 60 kt at or above the LRP and 0 below; 60/30/5/0 kt by stage at θ, 0.75θ, 0.5θ; 0 below the LRP then 60/90 kt (stage at 1.25θ) or 30/60/90 kt (stages at 1.25θ, 1.5θ)
H_t, R_t, P_t	J-17 annual mean head (ft), total recharge, total pumping (10^3 acre-ft)
$\alpha, \beta, \gamma, \delta$, $a = 1 + \delta$	affine map coefficients; $a = 0.7461$
$\underline{R}$	recharge floor: 43.7 (1956), 166.5 (q05), 179.1 (q10)
$\bar{P} = 282.16$	fit-window mean pumping (BAU)
$H^*(P)$	worst-floor attractor $(\alpha + \beta\underline{R} + \gamma P)/(1 - a)$
ρ^*	securing fraction of $\bar{P}$ for 618 ft under the worst floor (0.9278; cut 7.2%)
flat q%, S1, CPM	pumping q% of $\bar{P}$; -20% below 660 ft; $-20/-30/-35/-40$% below 660/650/640/630 ft